

**PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)**  
**ON**  
**AutoFix – Instant Vehicle Booking Service**  
**submitted in partial fulfillment of the requirements for the award of degree of**  
**BACHELOR OF ENGINEERING**  
**in**  
**COMPUTER SCIENCE AND ENGINEERING**

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## **DECLARATION**

This is to certify that the project report entitled “**AutoFix – Vehicle Service Booking and Management System**” submitted in partial fulfillment of the requirement for the award of the degree of Bachelor of Engineering in Computer Science and Engineering to the Department of Computer Science and Engineering, Chitkara University Institute of Engineering and Technology, Chitkara University, is a bonafide record of our original work under the guidance and supervision of **Dr. Rajat Takkar**.

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## **Abstract**

The Vehicle Service Booking and Management System, "AutoFix" project aims to make vehicle servicing more convenient and digitised with a single web-based platform. Traditionally, clients book services by calling or visiting service centres. This process is inefficient, time-consuming, difficult to schedule and not transparent, making it difficult for customers to monitor their vehicle service status. Their ability to schedule appointments, maintain records, and ensure smooth workflows to complete their work is also difficult. This makes the process more complex and increases their workload.

AutoFix helps fix these issues by providing car owners with a clean and efficient mobile service tool, which connects with vehicle service providers. Customers can create profiles and look at different vehicle services, such as oil changes and engine monitoring.

AutoFix solves these problems by providing a neat and user-friendly online system that links vehicle owners to service centres. Users can create accounts on the website and view different service options for their car, including oil changes and engine diagnostics. These issues are addressed by AutoFix, which provides an efficient, user-friendly digital solution that can connect car owners and service centres. Customers can register for accounts on the system, browse a variety of car services, such as oil change, engine performance check and general maintenance, and book appointments with local garages according to their availability. The use of a slot-based timetable helps prevent overbookings, minimising the wait time and improving efficiency.

The system allows customers to receive information on the status of their car servicing, and view their bookings through a user-friendly interface. On the service provider side, mechanics and managers have an admin dashboard. This allows them to view appointments, service status, availability, and generate billing information. Users, appointments, services and all other details are securely stored electronically.

This reduces the amount of physical documents and simplifies data management. The application uses the latest web technologies, including React.js for the user interface, Node.js/Express.js for the server, and MongoDB database. This stack of technologies ensures scalability, speed, and a good user experience.

# 1. Introduction

In today's fast moving world, vehicle owners expect easy and reliable services that save time and offer complete transparency. However, a large number of automobile service centers still use old methods such as phone calls, manual registers, and in-person visits to book service appointments. These methods are often inefficient, time-consuming and subject to scheduling conflicts. Customers often find it difficult to check service slots, track the progress of their car being serviced and keep a record of previous repairs & maintenance.

Service providers also have a number of operational challenges. Manually managing customer appointments can lead to double bookings, underutilization of resources and communication gaps between mechanics and customers. Maintaining service histories, job status changes and invoices on paper-based systems reduces administrative efficiency and increases the workload. These limitations become more significant as the number of customers increases and can result in customer dissatisfaction.

To solve these problems, AutoFix – Vehicle Service Booking and Management System is developed as a complete web based system that connects vehicle owners with service centers and mechanics. The system lets the customers to register, view services offered, select a convenient time slot and make bookings online. Users can also track the status of their vehicle servicing and can view their booking history. Simple and Intuitive interface.

AutoFix also provides the service provider with a dedicated administrative dashboard for use by mechanics and administrators. With this dashboard they can manage bookings, update the progress of services, set availability and generate invoices. The slot-based scheduling mechanism ensures the organisation and efficiency of service operations, and avoids overbooking.

AutoFix is built using the MERN stack with React and Tailwind CSS for the frontend, Node.js and Express.js for the backend, and MongoDB for storage. This modern technology stack makes the application scalable, secure, responsive and easy to maintain. The system also provides role-based authentication to ensure that users can only access features that are relevant to their roles.

## **1.1 Problem Background**

In today's fast-paced world, it's not always easy for car owners to get reliable and timely service for their vehicles. Many local garages still use manual systems, phone calls, or in-person bookings. This can lead to long waits, problems with scheduling, and not being able to get clear information about how their car repairs are going. Customers often have trouble finding reliable service centres, comparing the services they offer, or keeping track of the status of their car repairs. Mechanics and garage owners also have trouble keeping track of appointments, storing customer information, and making sure that service times don't overlap. There isn't an easy-to-use, all-in-one digital platform that connects car owners with service providers and lets them book services, keep track of how they're going, and manage their schedules. Because of this, there needs to be a smart and organized system like AutoFix to make it easier to book vehicle service and make the service as a whole work better.

## **1.2 Objective**

The main aim of the AutoFix – Vehicle Service Booking and Management System is as follows:

1. To build a web platform for booking vehicle services where customers can register, see services and book appointments online.
2. Provide a dashboard to manage bookings, service status and availability Automate appointment and service management for mechanics and administrators
3. To reduce manual work and avoid double bookings by introducing a slot-based booking system and centralizing digital record management.
4. To enable real time service tracking and booking history for customers to track the progress of vehicle servicing.
5. To develop a secure, scalable and user friendly system using MERN stack with role based authentication and responsive design for accessing across multiple devices.

## **2. Methodology**

The methodology adopted for the development of AutoFix – Vehicle Service Booking and Management System follows a structured approach to design, develop, and deploy a web-based platform for vehicle service booking and management. The system is developed using the MERN stack to ensure scalability, responsiveness, and efficient data handling.

### **Step 1: Requirement Analysis**

The problems associated with conventional vehicle service booking methods like manual appointment scheduling, lack of transparency, and poor record management were identified. Functional requirements were collected for customers, mechanics and administrators.

### **Step 2: System Design**

The system architecture and database schema were designed to define the relationships between users, services, bookings and invoices. System structure and interactions were illustrated by preparing UML diagrams such as Use Case Diagram, Class Diagram.

### **Step 3: Frontend Development**

The user interface was developed using React and Tailwind CSS. Responsive pages were created for registration, login, service browsing, booking appointments, and dashboard management.

### **Step 4: Backend Development**

The server-side logic was implemented using Node.js and Express.js. REST APIs were developed to handle authentication, booking management, service updates, and invoice generation.

### **Step 5: Database Integration**

MongoDB was used to store user details, vehicle information, service records, appointments, and billing data. The database provides secure and centralized storage for all application data.

## Step 6: Authentication and Role Management

Role-based access control was implemented to provide separate access for customers, mechanics, and administrators. Secure login and registration ensure that users can access only authorized features.

## Step 7: Testing and Deployment

The system was thoroughly tested to verify functionality, responsiveness, and security. After successful testing, the application was deployed for practical use and future enhancement.

## 3. Tools and Technologies

The **AutoFix – Vehicle Service Booking and Management System** uses a combination of modern web technologies to deliver a secure, scalable, and user-friendly solution for vehicle service booking and management.

- **React:** Used to build the interactive and responsive frontend user interface.
- **Tailwind CSS:** Used for styling the application and creating a modern, mobile-friendly design.
- **Node.js:** Used to implement the server-side logic and handle backend operations.
- **Express.js:** Used to develop RESTful APIs for authentication, booking, and service management.
- **MongoDB:** Used to store user information, appointments, service records, and invoice details.
- **Git & GitHub:** Used for version control, collaboration, and source code management.

The project follows the **MERN stack development approach**, where the frontend communicates with the backend through REST APIs, and all application data is stored in MongoDB. A slot-based scheduling mechanism is implemented to prevent overbooking and ensure efficient appointment management. Role-based authentication secures the system, while responsive design ensures compatibility across desktops, tablets, and smartphones.

These technologies work together to provide a reliable, scalable, and efficient platform that simplifies the vehicle servicing process for both customers and service providers.

## **4. Implementation**

AutoFix – Vehicle Service Booking and Management System has been designed and developed in a modular architecture to ensure maintainability, scalability and ease of future enhancement. The system consists of a number of independent modules, each of which is responsible for a certain set of functionalities.

### **1. User Login Module**

This module deals with user registration, login and secure authentication. It has role based access control for customers, mechanics and admin so each user can access only those features which are relevant to him.

### **2. Service Management Module**

This module enables the administrator to add, update and manage the various services such as oil changes, engine diagnostics and general maintenance for the vehicles.

### **3. Booking Module**

Customers can choose services, select available time slots and schedule appointments online. The slot based scheduling system prevents overbooking and helps in proper allocation of service times.

### **4. Admin Dashboard Module**

This dashboard is used by mechanics and administrators to manage appointments, update service progress, define availability and generate invoices.

### **5. Database Module**

MongoDB stores user data, appointment details, service information, and billing records in a centralized and secure manner.

## 5. Major Findings & Output

The introduction of AutoFix – Vehicle Service Booking and Management System led to substantial improvements in the process of servicing vehicles for both customers and service providers. The system successfully digitizes the traditional process of booking services, and provides a centralized platform for managing appointments, tracking service progress and maintaining service records.

One of the major findings of the project is that the manual effort is significantly reduced. Customers can book the vehicle services online, choose the available time slots and check the status of their service without visiting the service center or making repeated phone calls. This increases transparency throughout the servicing process, saves time and improves convenience.

It also enhances the operational efficiency for service providers. A dedicated admin dashboard lets mechanics and admins handle appointments, update service status, set availability, and create invoices. The slot-based scheduling mechanism is advantageous for avoiding double bookings and making the best use of the available service resources.

Also important is the better communication between customers and service centers. Customers can track the progress of their vehicles with real-time service status updates and centralized record management, leading to increased trust and customer satisfaction.

Modern technologies such as React, Node.js, Express.js and MongoDB are used to make the application responsive, scalable and reliable. Secure authentication and role-based access control provide additional data protection and system integrity.

The results of the project show that the system successfully provides the key interfaces such as landing page, login and signup pages, service page, booking page and admin dashboard. They offer a comprehensive and easy to use solution for booking and managing vehicle service together.

AutoFix streamlines operations, reduces scheduling errors, improves visibility and offers a working digital solution to bring vehicle service operations into the 21st century.

## 5.1 Output

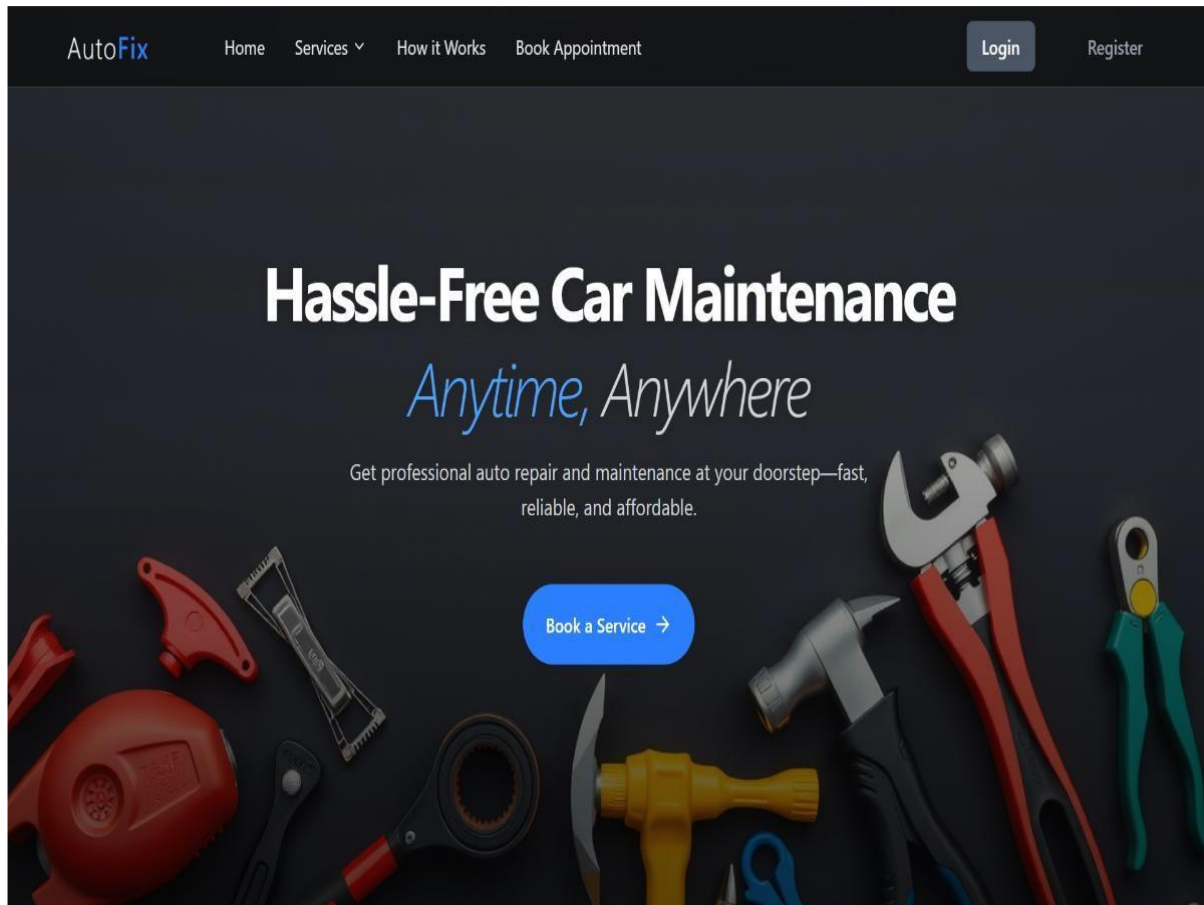


Figure 5.1.1

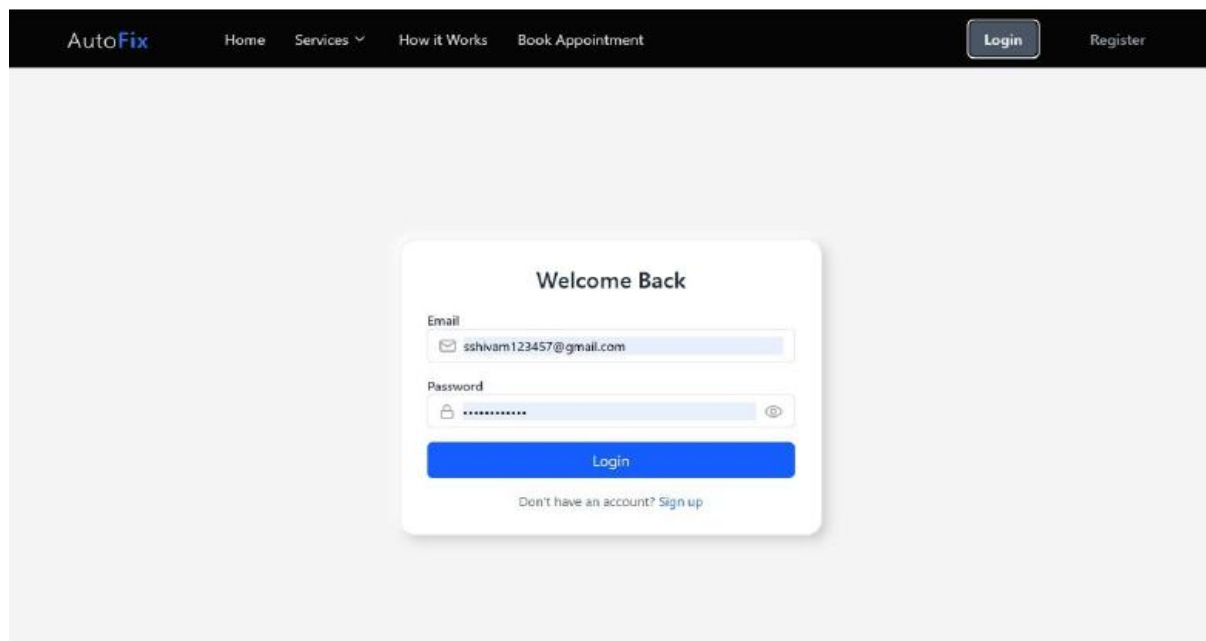


Figure 5.1.2

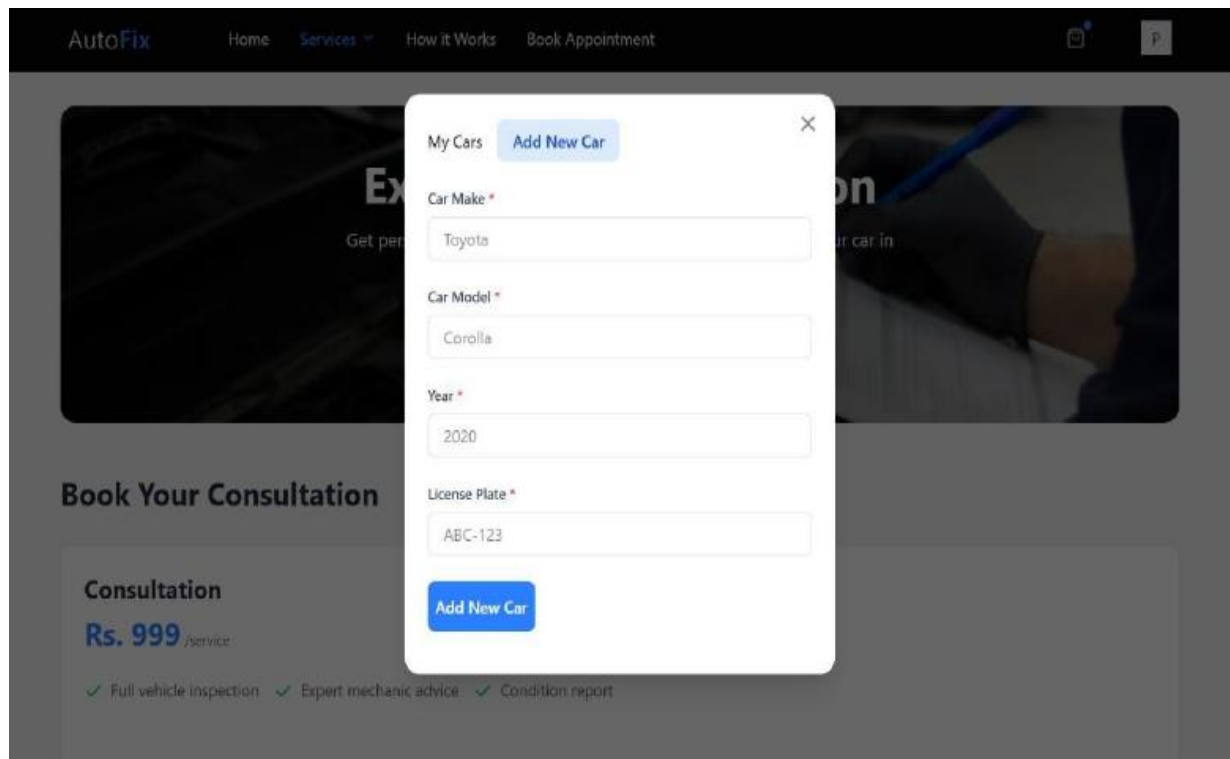


Figure 5.1.3

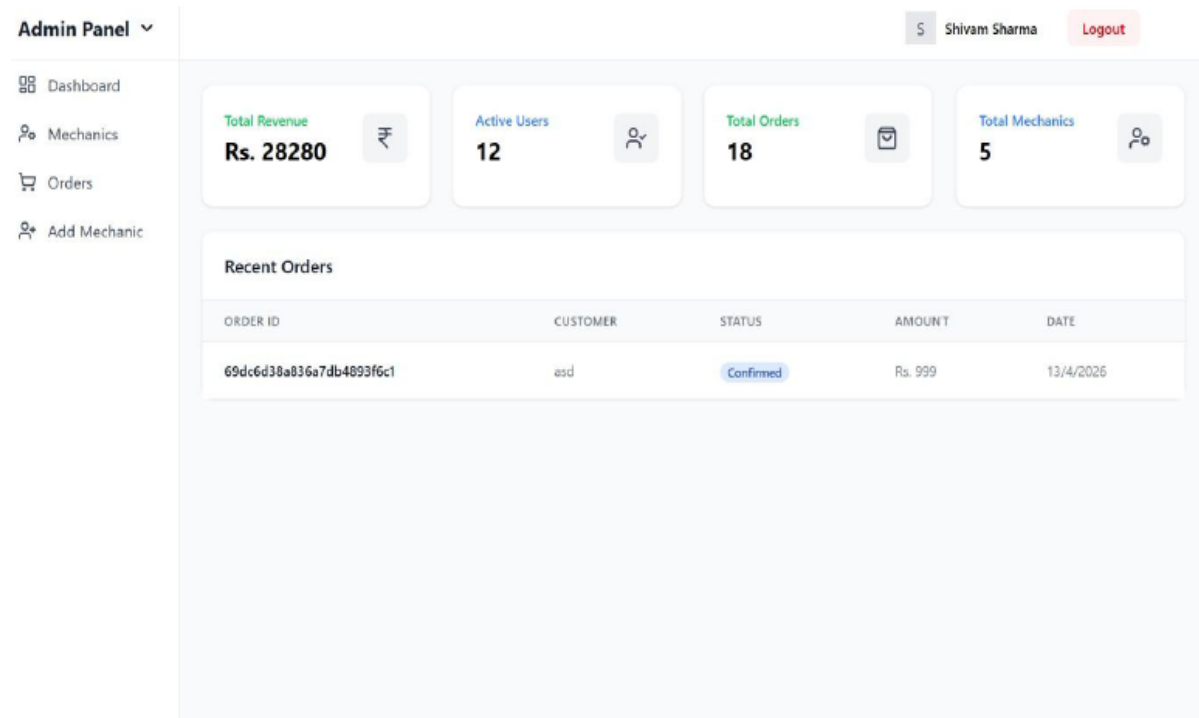


Figure 5.1.4

## 6. Conclusion

The AutoFix – Vehicle Service Booking and Management System showcases how modern web technology can be used to convert the traditional vehicle servicing process into a smart, efficient and user-friendly digital platform. This project deals with the major problems in the conventional service booking, such as manual appointment booking, lack of transparency, weak communication, and inefficient record keeping.

AutoFix provides customers with a unified web-based solution to register, search available services, book appointments and monitor the status of their vehicle servicing in real-time. There is also a separate dashboard for mechanics and administrators to manage bookings, update service progress, set availability and create invoices. This reduces the manual effort considerably and improves the operational efficiency.

The stack used, MERN stack (React, Node.js, Express.js and MongoDB) ensures that the system is scalable, responsive and maintainable. The platform's reliability and security are improved by role-based authentication and secure data storage, and responsive design ensures the application is accessible on desktops, tablets, and smartphones.

The project demonstrates the potential of digital solutions to improve customer satisfaction and to streamline service center operations. AutoFix builds a transparent and efficient ecosystem for vehicle service management by reducing scheduling conflicts, minimizing paperwork, and providing real-time updates.

In summary, AutoFix offers a practical and effective approach to modernising the booking and management of vehicle service. This improves the convenience of customers and service providers and also lays a solid foundation for future enhancements like online payments, push notifications, AI-based service recommendations, predictive maintenance alerts, and more. Overall the project is a demonstration of the power of technology to deliver a smarter and more efficient automotive service experience.

## 7. Future Scope

The AutoFix – Vehicle Service Booking and Management System provides a strong framework for digitizing vehicle service operations and can be enhanced with several advanced features to enhance functionality and user experience.

One of the major future enhancement is integration with online payment gateways like Razorpay, PayPal, Stripe etc. This would allow customers to make secure payments directly on the platform after booking services or receiving invoices.

The system can also be integrated with push notifications, SMS and Email alerts to keep the customers updated on booking confirmations, service progress and completion updates. Integration with messaging services would make communication easier and reduce the need for manual follow-up.

Also handy, is the inclusion of AI-based service recommendations and predictive maintenance alerts. It could also look at the vehicle's usage and service history and suggest maintenance services that might be appropriate as well as warn the user of upcoming service requirements.

A mobile application can be developed for android and iOS for providing more convenient experience for customers and service providers which will allow the customer and service provider to access the platform anytime and anywhere.

The platform can also include features such as customer reviews and ratings, analytics for service centers, and inventory management for spare parts and consumables. They would do this by providing service centers with the ability to improve operational efficiency and sustain higher service quality.

More integration with GPS and location services could help customers find nearby garages and provide navigation support. Multi-branch support can be implemented as well, to allow service providers to manage appointments for multiple locations.

In general, these future improvements will make AutoFix a full-fledged and intelligent vehicle service ecosystem, and make it more powerful, scalable and ready for the real world commercial deployment.

## 8. References

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- Node.js Official Documentation – <https://nodejs.org/en/docs>
- Express.js Official Documentation – <https://expressjs.com>
- MongoDB Official Documentation – <https://www.mongodb.com/docs>
- Git Official Documentation – <https://git-scm.com/doc>
- JavaScript Documentation – MDN Web Docs – <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
- REST API Design Best Practices – Microsoft Learn – <https://learn.microsoft.com/en-us/azure/architecture/best-practices/api-design>

## 9. Appendices

1. Sample screenshots of the Landing Page, Service Page, Login Page, Signup Page, Booking Page, and Admin Dashboard.
2. Code snippets of core modules including user authentication, service management, appointment booking, service tracking, and invoice generation.
3. System architecture diagrams illustrating the overall workflow and data flow of the application.
4. Use Case Diagram and Class Diagram representing system design and relationships between entities.
5. Sample MongoDB document structures for users, services, bookings, and invoices.
6. Example outputs showing appointment booking details, service status updates, and generated invoices.
7. Testing results and performance observations related to responsiveness, booking operations, and dashboard functionality.
8. User interaction examples demonstrating customer booking flow and admin dashboard operations.
9. Error handling cases and system behavior for invalid logins, booking conflicts, and unauthorized access.
10. Installation and setup instructions for running the AutoFix project locally using the MERN.